

A Posterior Model Projection of Narrow-domain ExoEarth Candidates in the Milky Way’s 7–9 kpc Annulus

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ABSTRACT

Kepler occurrence rates are defined per target star, whereas a Galactic projection additionally requires an explicit host-star measure and a transport assumption. We reconstruct the Bryson et al. Model 1 occurrence posterior from the public DR25 inputs using corrected two-sided measurement-error propagation, 400 reliability/measurement realizations per completeness branch, and adaptive ensemble Markov chain Monte Carlo. We propagate every retained posterior realization through a pinned Just–Jahreiß/PARSEC host population. The target is narrowly defined by $0.9 \leq R_p/R_\oplus \leq 1.1$ and $0.9 \leq S_p/S_\oplus \leq 1.1$, intersected with the present-day conservative habitable zone, around operationally main-sequence 5300–6000 K stars at least 4.57 Gyr old in the 7–9 kpc annulus. The host model gives $N_\star = 2.6306 \times 10^8$. The constant-completeness scenario yields median $\langle f_{\text{EE}} \rangle = 0.0123$ (68% interval 0.0058–0.0235) and $\Lambda_{\text{EE}} = 3.22$ million (1.52–6.19 million). The zero-completeness scenario yields 0.0174 (0.0080–0.0339) and 4.57 million (2.10–8.91 million), respectively. Both branches pass the declared convergence and Monte Carlo error gates. However, the exact nominal DR25 target contains zero candidates and zero summed catalog reliability, and more than 95% of corrected catalog realizations also contain none. The reported intervals therefore quantify the fitted posterior conditional on the adopted separable occurrence model and frozen host model; they do not include local support, occurrence-model-form, climate, host-population, or Galactic-transport uncertainty. The result is a reproducible conditional model projection, not a direct candidate-supported occurrence measurement and not a census of habitable or inhabited worlds.

Keywords: Exoplanet astronomy — Exoplanet population studies — Habitable planets — Milky Way disk — G dwarf stars — Astrostatistics

1. INTRODUCTION

The Kepler mission established that small planets are common and made occurrence-rate inference central to exoplanet demography (e.g., [A. W. Howard et al. 2012](#); [F. Fressin et al. 2013](#); [E. A. Petigura et al. 2013](#)). Modern analyses must account for target selection, injection–recovery completeness, catalog reliability, stellar-property uncertainties, and the functional form of the underlying population ([C. J. Burke et al. 2015](#); [D. Foreman-Mackey et al. 2014](#); [M. Kunitomo & J. M. Matthews 2020](#)). These issues are most severe for small planets at low instellation, where detections are sparse and smooth models are extrapolated into weakly supported regions.

[S. Bryson et al. \(2021\)](#) formulated a differential occurrence model in planet radius, instellation, and host effective temperature. Its output is a number of planets per star. Converting that result into a Galactic population expectation requires a second measure describing eligi-

ble stars and an assumption that the Kepler-calibrated occurrence law can be transported to that population.

Version 3 of this analysis implemented that transport with fixed median parameter vectors. The present work replaces those plug-in values with a newly reconstructed, corrected joint posterior; validates the host and TAMS selection; audits direct DR25 support in the exact target; and freezes numerical, model, climate, and spatial sensitivities before interpreting the result. The constant- and zero-completeness treatments remain separate scenarios and are never pooled into a confidence interval.

The demonstration intentionally answers a narrow astronomical question. It does not infer inhabited planets, biospheres, continuously habitable worlds, or complete physical Earth analogs. It estimates a radius–instellation selection around an old, operationally defined G-star population within a geometric Galactic annulus.

2. ESTIMAND AND NOMENCLATURE

We define dimensionless planet radius and bolometric instellation as

$$r \equiv \frac{R_p}{R_\oplus}, \quad I \equiv \frac{S_p}{S_\oplus}. \quad (1)$$

A *narrow-domain ExoEarth candidate* satisfies

$$0.9 \leq r \leq 1.1, \quad 0.9 \leq I \leq 1.1, \quad (2)$$

and lies in the present-day conservative circumstellar habitable zone of its host, evaluated with the $1 M_\oplus$ inner-boundary prescription of R. K. Kopparapu et al. (2013, 2014). The modifier “narrow-domain” is essential because ExoEarth has no unique radius convention across the literature.

This definition does not determine planet mass, density, composition, atmospheric state, surface water, geologic activity, magnetic field, or biology. Radius alone is not a sufficient compositional classifier (L. A. Rogers 2015), and present-day habitable-zone membership does not imply continuous residence in the zone.

Target hosts are thin- or thick-disk stellar assemblies satisfying

$$5300 \leq T_{\text{eff}} \leq 6000 \text{ K}, \quad t_\star \geq 4.57 \text{ Gyr}, \quad (3)$$

classified as main-sequence objects by Section 3.2, within

$$7 \leq R_{\text{GC}} \leq 9 \text{ kpc}. \quad (4)$$

$N_\star$ is the expected number of selected hosts. Λ_{HZ} is the expected number of planets with $0.5 \leq r \leq 1.5$ in the present-day conservative HZ, and Λ_{EE} is the expectation in Equation (2). These are expected planets, not the probability that a star hosts at least one planet. The result is conditional on

$$\mathcal{M} = \{M_{\text{JJ}}, M_{\text{iso}}, M_{\text{TAMS}}, M_{\text{occ}}, M_{\text{HZ}}, D_R, Q_R\}, \quad (5)$$

which denotes the Galactic model, isochrones, stellar selector, occurrence model, climate prescription, radial domain, and quadrature.

3. HOST-STAR POPULATION

3.1. Generalized Just–Jahreiß model

The host population is generated with the generalized Just–Jahreiß (JJ) Galactic disk model (A. Just & H. Jahreiß 2010; K. Sysoliatina & A. Just 2022), pinned to commit 2828a2e8bfc3 (jjmodel v1.0.1; the full hash is given in Section 11). We use the pinned `tutorial2/parameters` and `sfrd_peaks_parameters` realization, the default four-slope broken-power-law IMF (`imfkey=0`), a solar Galactocentric distance of 8.17

kpc, and a runtime radial grid $4 \leq R \leq 14$ kpc with $\Delta R = 0.5$ kpc. Only the thin and thick stellar disks are retained; the stellar halo is excluded.

Stellar assemblies are generated with the JJ-distributed Padova library based on PARSEC v1.2S tracks (A. Bressan et al. 2012) and the associated PARSEC–COLIBRI isochrone generation (P. Marigo et al. 2017); the workflow downloads the multiband archive identified by DOI 10.11588/data/ZCXHOE. The COLIBRI extension is part of the library provenance but does not determine the selected main-sequence G-star rows. Each synthetic row i at radial node R_j carries a vertically integrated present-day surface-number weight w_{ij} in pc^{-2} .

The resulting host count is a JJ/PARSEC-conditioned expectation, not a direct Gaia census. The model is axisymmetric and vertically integrated; spiral arms and azimuthal substructure are not resolved. The sub-percent radial-grid diagnostic reported below is not a posterior uncertainty on $N_\star$: uncertainties in the JJ parameter realization, IMF, age–metallicity and star-formation prescriptions, and isochrone family are not propagated in this work.

3.2. Operational main-sequence selector

For each synthetic row, the stellar radius is reconstructed from current mass and surface gravity,

$$\frac{R_{\star,ij}}{R_\odot} = \left[\frac{M_{f,ij}}{M_\odot} 10^{\log g_\odot - \log g_{ij}} \right]^{1/2}, \quad \log g_\odot = 4.438. \quad (6)$$

The host indicator is

$$\begin{aligned} H_{ij} = & \mathbf{1}(d_{ij} \in \{\text{thin}, \text{thick}\}) \mathbf{1}(5300 \leq T_{ij} \leq 6000) \\ & \times \mathbf{1}(t_{ij} \geq 4.57) \mathbf{1}(\log g_{ij} < 7) \\ & \times \mathbf{1}[R_{\star,ij} \leq R_{\text{TAMS}}(T_{ij})]. \end{aligned} \quad (7)$$

The final factor is the operational main-sequence classifier. The preceding $\log g < 7$ factor is only a veto against compact-remnant contamination, because a compact object can otherwise fall below a one-sided radius boundary.

$R_{\text{TAMS}}(T)$ is the frozen public `tams_parsec.txt` boundary from the Berger/Huber evolutionary-state implementation (T. A. Berger et al. 2018; D. Huber 2018). The exact upstream table is pinned at commit 5e904afad81805c4e3ac4c3f78510a2a1df33d14; the verbatim project copy has SHA-256 d2c47b264a298a599064a9e58f19f309886e7b96f36cc9603c9ca55494f87aac. Its generator uses solar-composition PARSEC v1.2S tracks ($Z = 0.017$, $Y = 0.279$) and reconstructs radius from luminosity and temperature. The canonical interpolation

is linear in T and $\log_{10}(R_{\text{TAMS}}/R_{\odot})$, and extrapolation is forbidden. Appendix C lists the adopted segment.

This is a temperature-dependent evolutionary boundary, not the fixed $R_{\star} < 1.35R_{\odot}$ cut rejected in the source analysis. It is used as a transport approximation to the Berger/Huber *Evo1* classification inherited by the Bryson denominator, not as a claim that the complete Kepler and JJ stellar selections are identical. A separate audit regenerated genuine low-mass solar-composition phase-7 nodes from 5151 to 6060 K. They reproduce the selected population exactly, including the 5300–5390 K interval, without using the table’s special 5200 K anchor. The archived metallicity-dependent differential test failed low-mass coverage and contamination checks and is excluded rather than reported as a numerical sensitivity.

Because the JJ age coordinate is a midpoint grid, the formal threshold 4.57 Gyr admits no row exactly at that age. The youngest selected midpoint is 4.5875 Gyr. A Stefan–Boltzmann radius reconstructed from L and T_{eff} differs from Equation (6) by 0.31% at the median and 0.40% at the 95th percentile. Only 0.336% of the integrated canonical host weight lies within 1% of the adopted TAMS boundary. Replacing the mass–gravity radius by the luminosity–temperature radius changes $N_{\star}$ by +0.131% and Λ_{EE} by +0.148% in the 7–9 kpc annulus, directly testing the classification stability of the radius reconstruction.

3.3. Host measure and radial quadrature

The selected-host surface density is

$$\Sigma_{\star}(R_j) = \sum_i H_{ij} w_{ij}. \quad (8)$$

The host count in the annulus is

$$N_{\star} = 10^6 \int_7^9 2\pi R \Sigma_{\star}(R) dR, \quad (9)$$

where R is in kpc and 10^6 converts kpc^2 to pc^2 . The canonical calculation evaluates this definition with the composite trapezoid rule on $R_j = (7, 7.5, 8, 8.5, 9)$ kpc,

$$N_{\star}^{(\Delta R)} = 10^6 \Delta R \sum_{j=0}^4 c_j 2\pi R_j \Sigma_{\star}(R_j),$$

$$c_0 = c_4 = \frac{1}{2}, \quad c_1 = c_2 = c_3 = 1. \quad (10)$$

with $\Delta R = 0.5$ kpc. Refinement to $\Delta R = 0.25$ kpc is treated as a numerical convergence check, not as an astrophysical uncertainty on the host normalization.

4. OCCURRENCE POSTERIOR AND HABITABLE-ZONE INTEGRATION

4.1. Bryson occurrence function

We adopt Model 1 from S. Bryson et al. (2021). The differential number of planets per star per unit r and I is

$$\lambda(r, I, T | \boldsymbol{\theta}) = F_0 C_1 r^{\alpha} I^{\beta} T^{\gamma} g(T), \quad (11)$$

where $\boldsymbol{\theta} = (F_0, \alpha, \beta, \gamma)$ is written in manuscript order, T is in kelvin, and

$$g(T) = \begin{cases} 10^{-11.839} T^{3.16}, & T \leq 5117 \text{ K}, \\ 10^{-16.769} T^{4.49}, & T > 5117 \text{ K}. \end{cases} \quad (12)$$

C_1 normalizes the model over $0.5 \leq r \leq 2.5$, $0.2 \leq I \leq 2.2$, and a uniform temperature measure over 3900–6300 K. Appendix A gives the analytic normalization.

The constant- and zero-completeness treatments are distinct likelihood scenarios inherited from the source analysis. They are not samples from one mixture distribution and are not assigned model probabilities here. The source interpretation places the unknown long-period completeness between these cases, but this statement is conditional on the adopted extrapolation family rather than an assumption-free physical bound.

4.2. Corrected catalog-error propagation

For a catalog value μ with reported lower and upper uncertainties σ_- and σ_+ , the v4 primary analysis draws

$$B \sim \text{Bernoulli}(1/2), \quad U = |Z|, \quad Z \sim \mathcal{N}(0, 1),$$

$$X = \begin{cases} \mu - \sigma_- U, & B = 0, \\ \mu + \sigma_+ U, & B = 1. \end{cases} \quad (13)$$

This construction places the quoted one-sigma lower and upper excursions at the 15.9th and 84.1st percentiles when the uncertainties are interpreted as two-sided Gaussian-equivalent errors. It replaces the source notebook’s signed-normal mixture, which does not preserve the intended association between direction and asymmetric scale. After perturbing instellation, radius, and T_{eff} , all source-domain conditions are reapplied:

$$0.2 \leq I \leq 2.2, \quad 0.5 \leq r \leq 2.5, \quad 3900 \leq T \leq 6300 \text{ K}. \quad (14)$$

The source-faithful legacy mechanism is retained only as a regression and sensitivity mode.

Each outer realization also applies the catalog reliability mechanism before fitting. Thus the calculation marginalizes over the implemented reliability and measurement procedure, but not over unmodeled catalog systematics or occurrence functional forms.

4.3. Poisson likelihood and adaptive MCMC

For retained candidates $x_p = (I_p, r_p, T_p)$, the source inhomogeneous-Poisson log likelihood can be written schematically as

$$\log \mathcal{L}(\boldsymbol{\theta}) = \sum_p \log \lambda(x_p | \boldsymbol{\theta}) - \int \lambda(x | \boldsymbol{\theta}) C(x) dx, \quad (15)$$

where $C(x)$ is the branch-specific summed completeness surface evaluated on the source grid. We retain the source uniform priors inside its numerical bounds: $0 \leq F_0 < 50000$, $-5 \leq \alpha < 5$, $-5 \leq \beta < 5$, and $-500 \leq \gamma < 50$, with zero density outside.

For each completeness branch we generate 400 independently seeded outer reliability/measurement realizations. Each four-parameter conditional posterior is sampled with 16 ensemble walkers after 1000 discarded burn-in steps. Autocorrelation time is checked every 1000 production steps. Acceptance requires, for every parameter, at least 100τ , no more than 5% relative change between successive τ estimates, and two consecutive qualifying checks. The constant production used a 20,000-step ceiling. Five zero-branch realizations reached that ceiling without satisfying the unchanged gate, so the entire zero branch was rerun with identical seeds and criteria and a 30,000-step ceiling; no failed realization was discarded.

Adaptive chains have unequal lengths. The aggregation therefore retains 1024 deterministically spaced post-burn rows from every outer realization, assigning equal weight to each catalog realization. Outer quantile Monte Carlo error is estimated by resampling whole realizations as clusters; matched contiguous chain blocks diagnose residual inner-chain error. The two completeness branches remain separate through this aggregation and the Galactic propagation.

4.4. Present-day conservative HZ

For a climatic boundary, the effective stellar flux is

$$S_{\text{eff}}(T) = S_{\text{eff},\odot} + aT' + bT'^2 + cT'^3 + dT'^4, \\ T' = T - 5780 \text{ K}, \quad (16)$$

using [R. K. Kopparapu et al. \(2013\)](#). The canonical conservative HZ uses the $1 M_{\oplus}$ runaway-greenhouse inner boundary from [R. K. Kopparapu et al. \(2014\)](#) and the maximum-greenhouse outer boundary. “Conservative” names this prescribed boundary pair; it is not a universal lower bound over climate models.

Let $S_{\text{out}}(T)$ and $S_{\text{in}}(T)$ denote the outer and inner flux boundaries. The broad HZ and narrow target domains are

$$\mathcal{D}_{\text{HZ}}(T) = [0.5, 1.5] \times [S_{\text{out}}(T), S_{\text{in}}(T)], \quad (17)$$

$$\mathcal{D}_{\text{EE}}(T) = [0.9, 1.1] \times [\max\{0.9, S_{\text{out}}(T)\}, \min\{1.1, S_{\text{in}}(T)\}], \quad (18)$$

with an empty domain when the upper limit does not exceed the lower. Across 5300–6000 K the outer HZ does not clip $I = 0.9$; the runaway-greenhouse boundary crosses $I = 1.1$ near 5727 K and clips the inner edge at lower temperatures.

4.5. Posterior Galactic estimator

For $k \in \{\text{HZ}, \text{EE}\}$ and posterior draw s , the expected planets per star at temperature T are

$$f_{k,s}(T) = \iint_{\mathcal{D}_k(T)} \lambda(r, I, T | \boldsymbol{\theta}_s) dr dI. \quad (19)$$

At radial node R_j , the occurrence-weighted surface density is

$$\Sigma_{k,s}(R_j) = \sum_i H_{ij} w_{ij} f_{k,s}(T_{ij}). \quad (20)$$

The propagated estimator is

$$\Lambda_{k,s}^{(\Delta R)} = 10^6 \Delta R \sum_{j=0}^4 c_j, 2\pi R_j \sum_i H_{ij} w_{ij} f_{k,s}(T_{ij}). \quad (21)$$

Posterior summaries are quantiles of $\{\Lambda_{k,s}\}$ after equalizing the outer-realization mixture. Equation (21) preserves the temperature dependence of both occurrence and HZ clipping; it is not a single rate multiplied by a host count.

5. RESULTS

5.1. Host population

The canonical selector yields

$$N_{\star} = 263,061,992.37 \quad (22)$$

model hosts in the 7–9 kpc annulus. The thin disk contributes 2.11×10^8 and the thick disk 5.23×10^7 , or 19.9% of the total. From the pre-TAMS parent measure of 2.9877×10^8 , the radius boundary rejects 3.3857×10^7 and the compact-object veto rejects 1.8513×10^6 in the mutually exclusive audit decomposition. Table 1 summarizes the selected host distribution.

The genuine solar-composition low-mass phase-7 curve selects exactly the same integrated population as the inherited one-dimensional boundary. In particular, omitting the special 5200 K table anchor changes $N_{\star}$ and the plug-in Λ_{EE} by 0.0000%. This validates the numerical implementation but does not quantify uncertainty from metallicity, helium abundance, or isochrone-family choice.

Table 1. Integrated host-property distribution in the 7–9 kpc annulus

Quantity	Unit	Weighted mean	16th percentile	Median	84th percentile
Effective temperature	K	5646	5424	5670	5841
Age	Gyr	9.35	6.41	9.59	12.16
[Fe/H]	dex	-0.020	-0.284	+0.033	+0.222
Current mass	$M_{\odot}$	0.913	0.810	0.910	1.015
Stellar radius	$R_{\odot}$	1.035	0.864	0.999	1.237
$\log g$	dex (cgs)	4.379	4.250	4.400	4.492

5.2. Occurrence and Galactic expectations

Table 2 gives the frozen posterior quantiles. Under constant completeness, the median narrow-domain occurrence is 0.01226 planets per selected star and the propagated median is

$$\Lambda_{\text{EE}}^{\text{const}} = 3.224 \times 10^6, \quad 68\% : [1.522, 6.189] \times 10^6. \quad (23)$$

Under zero completeness,

$$\Lambda_{\text{EE}}^{\text{zero}} = 4.572 \times 10^6, \quad 68\% : [2.103, 8.912] \times 10^6. \quad (24)$$

The corresponding 95% intervals are 0.652–10.842 million and 0.888–15.840 million. The branch medians differ by 41.83%, but that difference is a model scenario contrast rather than a credible-interval component.

Within matched posterior draws, the narrow target contains a median 3.119% of broad-HZ occurrence in the constant branch and 2.606% in the zero branch. This ratio is derived from the same occurrence model; it is not an independent planet fraction.

6. VERIFICATION AND EMPIRICAL-SUPPORT AUDIT

6.1. Statistical and numerical gates

All 400 constant-branch and all 400 zero-branch outer realizations passed the adaptive convergence gate. The minimum, median, and maximum production lengths were 8000, 12,000, and 17,000 steps for the constant branch and 9000, 14,000, and 22,000 for the zero branch. The minimum effective sample size over every parameter and realization was 1682.3 and 1662.4, respectively. There were no optimizer failures.

The largest outer-realization median Monte Carlo standard error, expressed as a fraction of the associated 16th–84th percentile width, was 1.63% for the constant branch and 1.56% for zero completeness. The corresponding maximum inner-chain fractions were 0.10% and 0.15%. Independent MCMC-seed families passed the frozen stability comparison. These values are far below the predeclared 10% outer and 5% inner gates.

The measurement implementation has a bitwise source-faithful legacy regression mode and a corrected

quantile-recovery suite. Aggregation rejects mixed modes, unequal outer weights, failed convergence, and inconsistent diagnostic metadata. The host/HZ propagation separately verifies non-negative weights, nested domains, unit conversions, and reproduction of the frozen quantiles. Refining the radial grid from 0.5 to 0.25 kpc changes the plug-in Λ_{EE} by +0.231%, below the predeclared 1% numerical threshold.

These checks establish engineering and numerical consistency of the frozen pipeline. They do not establish that the separable occurrence surface is an adequate scientific description inside the target box.

6.2. Direct DR25 support

The exact target is geometrically contained in the fitted DR25 source domain: $0.9\text{--}1.1 R_{\oplus}$ lies inside $0.5\text{--}2.5 R_{\oplus}$, $0.9\text{--}1.1 S_{\oplus}$ lies inside $0.2\text{--}2.2 S_{\oplus}$, and the selected temperatures lie inside 3900–6300 K. Geometric containment, however, does not guarantee local data support.

The merged nominal planet-candidate catalog contains 2277 rows. Fifty-four rows lie in the fitted source domain after the nominal cuts, with summed total reliability 42.23. For the selected 5300–6000 K stars, 54 candidates satisfy the narrow radius range, while only three satisfy the conservative-HZ instellation intersection. No nominal candidate satisfies the radius and fixed $0.9\text{--}1.1$ instellation box simultaneously; the exact climate-clipped target therefore has candidate count zero and reliability sum zero.

Across 400 corrected outer realizations, the constant branch has target-count quantiles (0, 0, 0, 0, 1) at probabilities (2.5, 16, 50, 84, 97.5)% and a 95.75% zero-count fraction. The zero branch has the same quantile vector and a 96.75% zero-count fraction. Only K08107.01 and K08242.01 enter the target under any perturbation, with 17 total appearances in the constant branch and 13 in the zero branch. Consequently, the local empirical-support gate is a scientific *failure*, even though the geometry and code checks pass.

Table 2. Frozen posterior occurrence and Galactic population quantiles

Branch	Quantity	2.5%	16%	Median	84%	97.5%
Constant	$\langle f_{\text{HZ}} \rangle$	0.0645	0.1601	0.3883	0.9070	1.9967
Constant	$\langle f_{\text{EE}} \rangle$	0.00248	0.00579	0.01226	0.02353	0.04121
Constant	$\Lambda_{\text{HZ}} (10^6)$	16.971	42.104	102.148	238.593	525.246
Constant	$\Lambda_{\text{EE}} (10^6)$	0.652	1.522	3.224	6.189	10.842
Zero	$\langle f_{\text{HZ}} \rangle$	0.0965	0.2537	0.6591	1.6216	3.7502
Zero	$\langle f_{\text{EE}} \rangle$	0.00338	0.00799	0.01738	0.03388	0.06021
Zero	$\Lambda_{\text{HZ}} (10^6)$	25.394	66.746	173.376	426.579	986.540
Zero	$\Lambda_{\text{EE}} (10^6)$	0.888	2.103	4.572	8.912	15.840

NOTE—Occurrence quantities are planets per star, not fractions of stars. Intervals are conditional on the fitted Model 1 occurrence surface and the frozen host/HZ construction.

Corrected v4 posterior: median, 68%, and 95% intervals

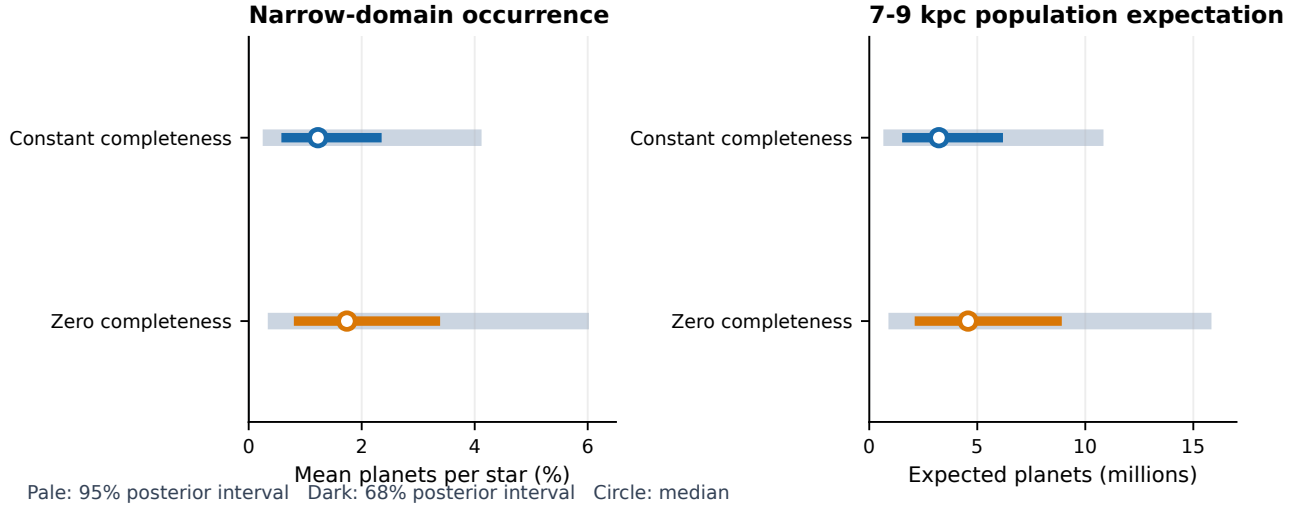


Figure 1. Corrected posterior summaries for narrow-domain occurrence per star (left) and the propagated 7–9 kpc population expectation (right). Pale and dark segments show 95% and 68% posterior intervals; circles mark medians. The two completeness branches are separate scenarios. These intervals do not include occurrence-model-form, local-support, host-population, climate, or transport uncertainty.

7. SOURCE-MODEL FIDELITY BENCHMARK

S. Bryson et al. (2021) report broad conservative-HZ occurrence of 0.37 planets per star under high completeness and 0.60 under low completeness for their broader $0.5\text{--}1.5 R_{\oplus}$, 4800–6300 K definition. The reconstructed v4 broad-HZ medians are 0.388 and 0.659. The agreement in scale supports source fidelity, but it is not independent validation: the present posterior is reconstructed from the same public DR25 model and inputs.

The point-vector reference inherited from version 3 gives $\Lambda_{\text{EE}} = 3.3765 \times 10^6$ for the constant branch. The corrected posterior median is 3.2238×10^6 ; these are different statistical functionals and need not coincide. The v4 inference uses the pooled equal-weight joint posterior

rather than independently sampled marginal summaries or a scaled broad-HZ interval.

8. SENSITIVITY AND UNCERTAINTY

The frozen register separates posterior scenarios, numerical checks, model alternatives, climate prescriptions, alternative estimands, categorical validation failures, and unparameterized systematics. These entries are not exchangeable random errors and are not combined in quadrature.

The legacy measurement algorithm changes the constant posterior median by only 0.026%, so correcting its asymmetric-error construction is methodologically necessary but does not drive the headline median. The host selector and climate prescription have much larger

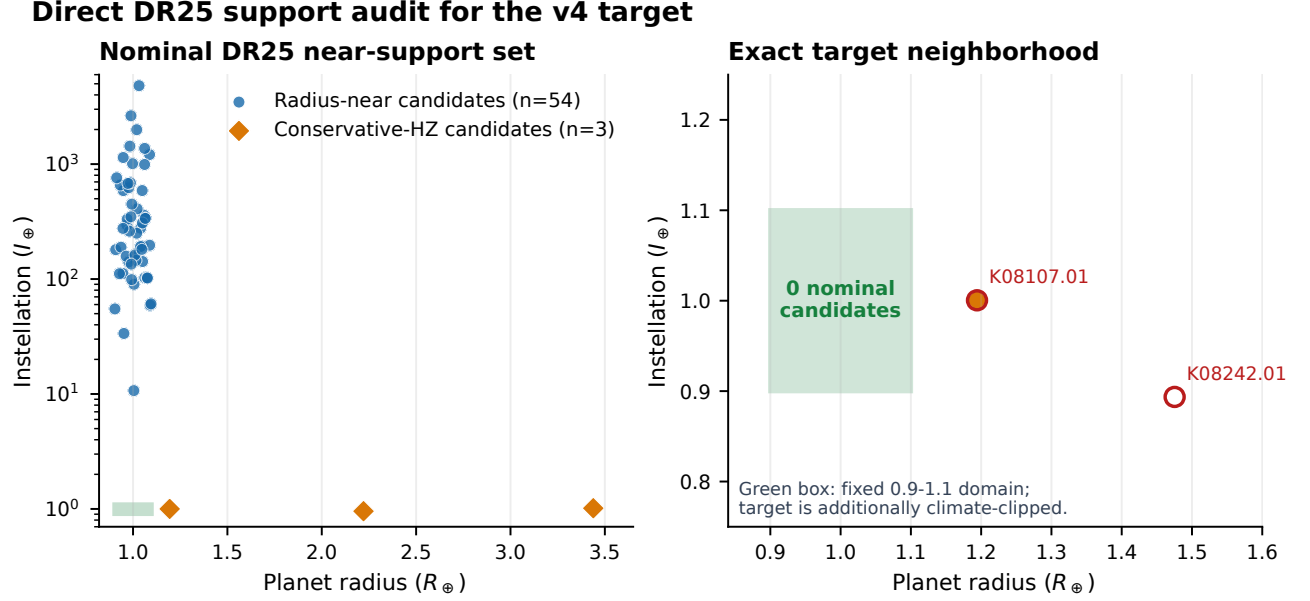


Figure 2. Nominal DR25 candidates near one component of the target selection. Left: radius-near and conservative-HZ-near candidates for 5300–6000 K hosts; the fixed target box is green. Right: the exact neighborhood, with the two nominal candidates that can enter only after measurement perturbation outlined in red. The climate condition further clips the fixed green box. No nominal candidate lies in the full target.

Table 3. Audited sensitivity of the narrow ExoEarth expectation

Category	Comparison	Change in Λ_{EE}	Statistical role
Measurement	Legacy versus corrected propagation	+0.026%	Posterior q50 regression
Completeness	Zero versus constant branch	+41.83%	Separate scenario
Host selector	Legacy fixed-log g , constant	-28.58%	Posterior model sensitivity
Host selector	Legacy fixed-log g , zero	-29.58%	Posterior model sensitivity
TAMS	Native curve without 5200 K anchor	0.000%	Plug-in numerical validation
Numerics	Radial grid 0.5 to 0.25 kpc	+0.231%	Plug-in convergence check
Occurrence	Bryson Model 2 versus Model 1	+3.30%	Point-vector model alternative
Climate	Inner HZ flux $\times 0.95$	-24.84%	Boundary perturbation
Climate	Inner HZ flux $\times 1.05$	+7.15%	Boundary perturbation
Climate	$0.1 M_{\oplus}$ runaway boundary	-57.28%	Alternative prescription
Climate	$5 M_{\oplus}$ runaway boundary	+7.15%	Alternative prescription
HZ definition	Optimistic versus conservative	+7.15%	Alternative estimand
Temperature	JJ weighted versus uniform	+0.380%	Weighting diagnostic

NOTE—The basis differs by row and is stated explicitly. Posterior rows use branch-specific medians; point-vector rows use the frozen constant reference 3.3765×10^6 . None is an added posterior error bar.

effects. Model 2 changes the point estimate by 3.30%, but no Model 2 posterior was propagated and that single alternative does not span arbitrary radius–installation structure.

The $\pm 1\%$ inner-boundary perturbations are asymmetric: multiplying the runaway-greenhouse flux by 0.99 changes the plug-in result by -3.67%, whereas a factor 1.01 gives +2.71%. Factors 0.95 and 1.05 give -24.84%

and +7.15%. This nonlinearity follows from clipping the fixed 0.9–1.1 installation box by the temperature-dependent inner HZ. The 0.1 and $5 M_{\oplus}$ rows alter the published climate prescription; they do not model the unknown planet-mass distribution inside the radius interval.

Changing the radial mask changes the estimand. Relative to the 7–9 kpc constant point-vector result, the

Frozen v4 sensitivity register

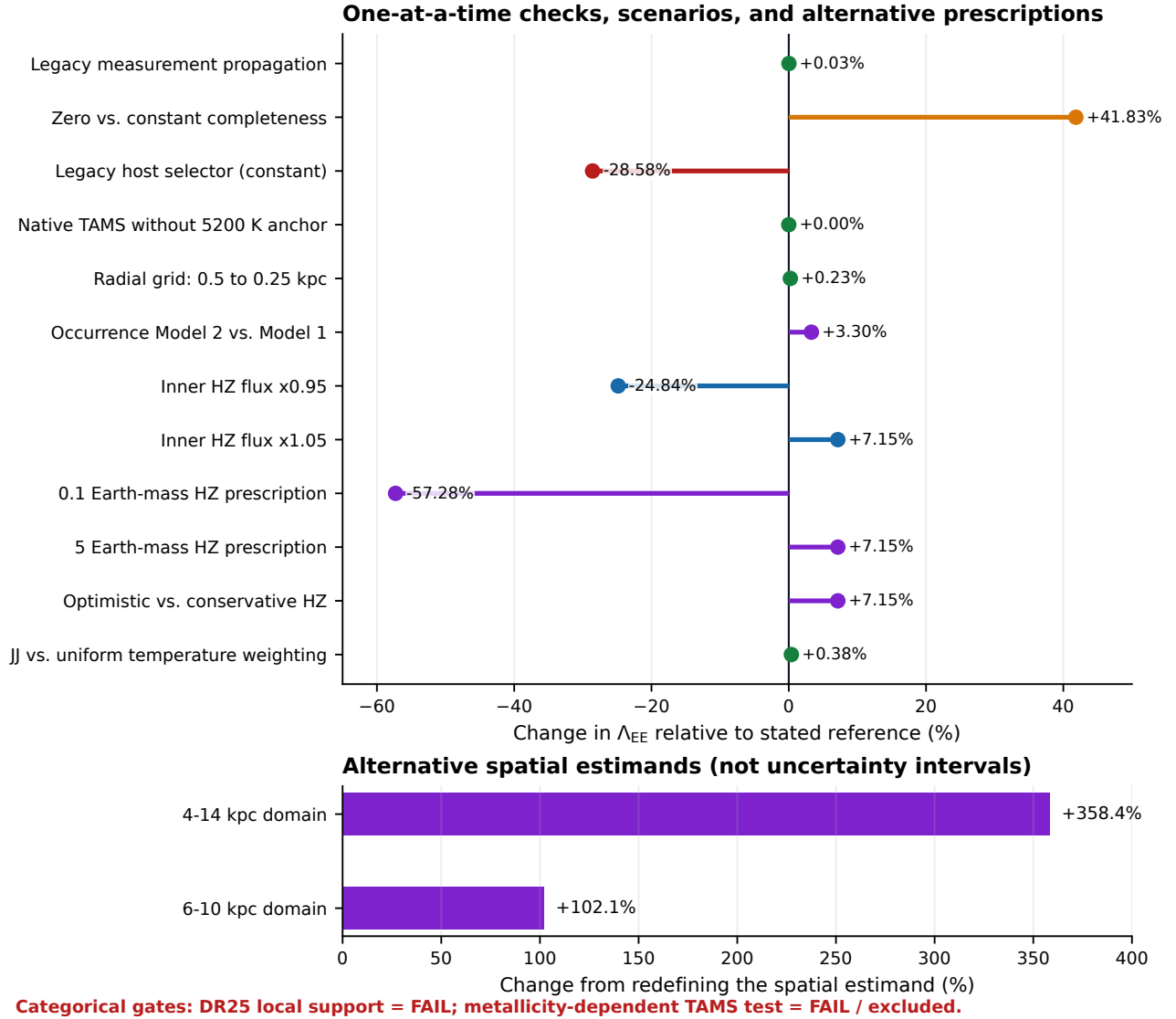


Figure 3. Frozen one-at-a-time sensitivity register. Colors distinguish numerical checks, scenarios, model alternatives, and alternative prescriptions. Spatial-domain changes are shown separately because they redefine the target population. The DR25 local-support failure and excluded metallicity-TAMS test are categorical findings, not plotted numerical error bars.

6–10 kpc domain gives $\Lambda_{EE} = 6.8239 \times 10^6$ (+102.10%) and the full 4–14 kpc JJ domain gives 1.5479×10^7 (+358.43%). They are alternative populations rather than uncertainty intervals for the annulus.

The archived metallicity-dependent TAMS surface is not a valid sensitivity. Its input contains 42 giant or high-mass phase-7 contaminants, reaching $155 M_{\odot}$ and $2312 R_{\odot}$, while several low-metallicity tracks do not cover 5300–6000 K. The previously reported +1.59% change is retracted and excluded. A valid low-mass metallicity-dependent TAMS analysis remains open.

Other unparameterized epistemic uncertainties remain open: JJ component normalization and star-

formation history, isochrone family, radial migration, multiplicity mismatch, occurrence transport with age and metallicity, and occurrence functional form in the locally empty target. No defensible joint distribution is available, so a combined total uncertainty is not reported.

9. DISCUSSION

9.1. Comparison with primary occurrence studies

Published Earth-analog occurrence estimates are not interchangeable because their radius, orbital coordinate, HZ definition, stellar sample, catalog, and reported statistical functional differ. The v4 median per-star oc-

currence, 1.23% under constant completeness and 1.74% under zero completeness, is numerically below broad-HZ estimates as expected for its much narrower integrated phase space.

S. Bryson et al. (2021) use $0.5\text{--}1.5 R_{\oplus}$, a complete conservative HZ, and 4800–6300 K stars for their quoted $\eta_{\oplus}$ values. Their 0.37 and 0.60 broad-HZ results are the source-model anchors for this analysis, not independent corroboration. M. Kunimoto & J. M. Matthews (2020) report an 84.1% upper limit below 0.18 planets per G star for $0.75\text{--}1.5 R_{\oplus}$ and 0.99–1.70 au; semimajor axis, radius, and host definitions all differ from v4. The conceptually narrower $\zeta_{1.0}$ of C. J. Burke et al. (2015) uses planet radius and orbital period within 20% of Earth, yielding a central value near 0.1 but an allowed 0.01–2 range once major statistical and systematic uncertainty is considered.

Using a stripped-core-aware occurrence interpretation, I. Pascucci et al. (2019) obtain broad-HZ values of roughly 5–10% and show that extrapolation choices can change the result by factors of several. G. J. Bergsten et al. (2022) report $\Gamma_{\oplus} = 15^{+6}_{-4}\%$, a local differential density per logarithmic period–radius area rather than an integrated planets-per-star quantity for the v4 box. Converting it would require an explicit domain and occurrence model.

No one of these primary results simultaneously matches the v4 radius, instellation, climate intersection, host-temperature range, and catalog/model definition. Point-estimate differences therefore do not by themselves establish inconsistency. The common conclusion relevant here is that sparse long-period data and occurrence-model form are material epistemic limitations.

9.2. Local support and model-form risk

The zero-candidate audit sharpens the limitation that version 3 described only as sparsity. The posterior remains mathematically finite because a separable power law is fitted over the larger source domain and then integrated over the target. Thus the interval describes uncertainty in that fitted projection; it does not demonstrate that the local surface is smooth, separable, or free of a break near Earth radius and instellation.

The Model 2 point comparison is useful but insufficient. It changes one temperature parameterization within the Bryson framework and does not test arbitrary curvature, radius–instellation interactions, or stripped-core structure (E. D. Lopez & K. Rice 2018; I. Pascucci et al. 2019). A more complete analysis would fit competing occurrence surfaces to the same perturbed catalogs

and perform posterior predictive checks targeted at the empty local region.

9.3. Host-population and transport uncertainty

The JJ host count is a deterministic output of the pinned parameter realization and isochrone library. The sub-percent radial-grid change is only a numerical diagnostic. Posterior uncertainty in star-formation and age–metallicity histories, IMF parameters, component normalizations, radial migration, and the choice among PARSEC, MIST, and BaSTI is not propagated. The reported host count is reproducible but not astrophysically known to sub-percent precision.

Equation (21) transports a Kepler-calibrated model to an older Galactic population. Conditional on T_{eff} , it assumes no additional dependence of planet occurrence or survival on age, $[\text{Fe}/\text{H}]$, $[\alpha/\text{Fe}]$, disk membership, or Galactocentric radius. This is a null transport assumption, not an empirical result.

The Kepler source denominator also includes observational-quality and binary cuts that the intrinsic JJ single-star population does not reproduce. Close companions can suppress planet occurrence (A. L. Kraus et al. 2016), while unresolved multiplicity alters radii and completeness. The direction and magnitude of a correction cannot be identified from the current inputs, so multiplicity remains unquantified rather than receiving an ad hoc factor.

9.4. Meaning of the Galactic annulus

The 7–9 kpc interval is an absolute geometric mask motivated by historically highlighted Galactic habitable-zone work (C. H. Lineweaver et al. 2004); it does not import that study’s time-dependent metallicity, supernova, terrestrial-planet, or biological weighting. Sharply bounded Galactic habitable zones are themselves model-dependent (N. Prantzos 2008; M. G. Gowanlock et al. 2011). The present result is therefore an estimate within an annulus, not the number of ExoEarth candidates in a physically complete Galactic habitable zone.

9.5. What the reported values mean

The statement $\Lambda_{\text{EE}} = 3.22$ million in the constant branch means that the frozen model stack assigns that median expected number of planets to the stated radius–instellation–present-day-HZ domain and host population. The zero branch assigns 4.57 million under a different completeness extrapolation. These are not observed counts, not numbers of systems with at least one selected planet, and not counts of rocky, ocean-bearing, habitable, or inhabited worlds.

The displayed posterior intervals are statistically meaningful for the implemented conditional model, but narrower in scope than the total scientific uncertainty. Reproducibility of a conditional projection and empirical validity of the projection are separate questions; v4 passes the former and explicitly fails the local-support component of the latter.

10. CONCLUSIONS

We reconstructed and propagated a corrected joint occurrence posterior through a temperature-resolved Galactic host measure. For the specified old G-star population in the 7–9 kpc annulus, the frozen host model gives $N_\star = 2.6306 \times 10^8$. The narrow-domain posterior results are

$$\begin{aligned}\Lambda_{\text{EE}}^{\text{const}} &= 3.22^{+2.97}_{-1.70} \text{ million,} \\ \Lambda_{\text{EE}}^{\text{zero}} &= 4.57^{+4.34}_{-2.47} \text{ million,}\end{aligned}\quad (25)$$

where uncertainties are 16th–84th percentile posterior intervals within each separate completeness scenario.

All 800 outer MCMC realizations pass the adaptive convergence gate, the outer and inner quantile Monte Carlo errors are small relative to posterior width, the radial quadrature is stable below 0.3%, and the native low-mass solar TAMS curve reproduces the canonical host selection. The legacy measurement mechanism changes the constant median by only 0.026%.

The main scientific limitation is not numerical. There are zero nominal DR25 candidates and zero summed reliability in the exact target, and more than 95% of corrected catalog realizations also contain zero. Occurrence-model form, host normalization, climate prescription, stellar multiplicity, and Galactic transport are not contained in the posterior intervals. The defensible conclusion is therefore conditional: the fitted separable Bryson Model 1 surface, transported through the

frozen JJ/PARSEC/TAMS population, projects a median of several million narrow-domain candidates into this annulus. The work does not provide a direct candidate-supported measurement or a complete astrophysical uncertainty budget.

11. DATA AND SOFTWARE AVAILABILITY

The public project repository is <https://github.com/jerroman/are-we-alone-in-the-universe>. The persistent project concept DOI is [10.5281/zenodo.20474527](https://doi.org/10.5281/zenodo.20474527); exact reproduction additionally requires the v4 release record and commit identified in its checksum manifest.

The immutable manuscript-v3 source is commit `5a3528aea6d6f28da8e9db4d40f0c84cbb43d501`. The v4 source-provenance manifest records the source tree and every imported LaTeX/BibTeX blob before editing. The Bryson research-code source is `d200f54b6f0df49e0dae530e69983cdce5397bfb`. Corrected-posterior execution is recorded by GitHub Actions runs `32470830404`, `32472776218`, `32506666772`, and `32527877921`. The release evidence contains the frozen numerical, host/TAMS, DR25-support, sensitivity, and primary-literature records together with SHA-256 manifests.

The analysis uses Python, NumPy (C. R. Harris et al. 2020), pandas (W. McKinney 2010), Matplotlib (J. D. Hunter 2007), Astropy (Astropy Collaboration et al. 2022), emcee (D. Foreman-Mackey et al. 2013), and jjmodel (K. Sysoliatina & A. Just 2022). The manuscript figures are generated only from committed frozen CSV/JSON artifacts. Local raw production artifacts are represented by their recorded hashes and GitHub Actions run identifiers; the reproducibility archive supplies the compact frozen outputs required to audit every reported number.

APPENDIX

A. ANALYTIC OCCURRENCE INTEGRALS

For exponent q , define

$$\mathcal{J}(x_0, x_1; q) = \begin{cases} \frac{x_1^{q+1} - x_0^{q+1}}{q+1}, & q \neq -1, \\ \ln(x_1/x_0), & q = -1. \end{cases} \quad (\text{A1})$$

With $T_0 = 3900$ K, $T_b = 5117$ K, and $T_1 = 6300$ K, the Model 1 normalization is

$$C_1^{-1} = \mathcal{J}(0.5, 2.5; \alpha) \mathcal{J}(0.2, 2.2; \beta) \times \frac{1}{T_1 - T_0} [10^{-11.839} \mathcal{J}(T_0, T_b; \gamma + 3.16) + 10^{-16.769} \mathcal{J}(T_b, T_1; \gamma + 4.49)]. \quad (\text{A2})$$

Because Equation (11) is separable within each temperature branch, $f_k(T)$ is evaluated analytically in r and I using Equation (A1). The only numerical quadrature in the final Galactic estimator is the radial integration after the row-level sums have been constructed.

B. BRYSON MODEL 2 SENSITIVITY

For the Model 2 sensitivity, the source occurrence function is

$$\lambda_2(r, I, T \mid \boldsymbol{\theta}_2) = F_0 C_2 r^\alpha I^\beta T^\gamma, \quad (\text{A3})$$

without the fixed geometric factor $g(T)$. Its normalization is

$$C_2^{-1} = \mathcal{J}(0.5, 2.5; \alpha) \mathcal{J}(0.2, 2.2; \beta) \frac{\mathcal{J}(T_0, T_1; \gamma)}{T_1 - T_0}. \quad (\text{A4})$$

The adopted [S. Bryson et al. \(2021\)](#) `hab2`, Poisson, “With Uncertainty,” constant-completeness median vector is

$$(F_0, \alpha, \beta, \gamma, C_2) = (1.13, -1.13, -0.85, +1.19, 1.072724 \times 10^{-5}). \quad (\text{A5})$$

Evaluated on the same frozen JJ rows and HZ prescription, this gives $\Lambda_{\text{EE}} = 3.487759 \times 10^6$, or +3.296% relative to the Model 1 constant-completeness reference.

C. FROZEN AND NATIVE PARSEC-TAMS BOUNDARIES

Table 4 lists the inherited monotonic segment used by the canonical 5300–6000 K selector. The 5200 K, $1.15R_\odot$ row is retained verbatim as source provenance. An independent regeneration uses genuine solar-composition low-mass phase-7 nodes at 5151.34 K ($0.75 M_\odot$), 5304.81 K ($0.80 M_\odot$), and 5390.14 K ($0.85 M_\odot$), followed by the native hotter sequence. This curve brackets the full target without the 5200 K row and produces no integrated selector disagreement. Thus the special anchor is not numerically responsible for the canonical host count.

Table 4. Frozen TAMS nodes used for G-star interpolation

T_{eff} (K)	$R_{\text{TAMS}}/R_\odot$
5200.00000	1.15000
5390.13944	1.22926
5517.85139	1.28542
5633.13293	1.35053
5738.25706	1.42375
5844.13178	1.49188
5951.82290	1.55332
6060.24246	1.61155

D. FROZEN VERIFICATION CHECKLIST

The v4 freeze requires: (1) input and artifact checksums; (2) bitwise legacy measurement regression and corrected quantile recovery; (3) post-perturbation source-domain filtering; (4) 400 of 400 converged adaptive chains in each branch; (5) equal outer-realization weights; (6) whole-realization outer MCSE and contiguous-block inner MCSE below their declared thresholds; (7) independent MCMC-seed stability; (8) finite non-negative host weights and occurrence

values; (9) radial-node, unit, and thin/thick closure; (10) correct temperature, age, component, TAMS, and compact-remnant selection; (11) nested HZ/target domains and analytic occurrence integrals; (12) native solar TAMS agreement without the 5200 K anchor; (13) 0.5-to-0.25 kpc radial convergence; (14) exact DR25 target-support masks; and (15) estimand-aware separation of posterior scenarios, point alternatives, and categorical failures.

The metallicity-dependent TAMS artifact does not pass this checklist. Its high-mass/giant contamination and missing low-metallicity temperature coverage are recorded as a failed, excluded experiment. Likewise, the zero-candidate DR25 result is a scientific support failure, not a software failure. These distinctions prevent engineering reproducibility from being mislabeled as observational validation.

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PERSONAL PHILOSOPHICAL OPINION

Regardless of how extensive and precise the data collected by our successors in the future may be, the equations with which they attempt to understand the universe will most likely remain primarily research models for estimating the expected number of worlds. It is difficult to expect them to provide an exact inventory of the actual state of affairs. With them, we will be able to connect data, test our assumptions, and determine the range within which the actual number of worlds most likely lies. Yet because of the limitations of observation, we will probably never arrive at a completely precise answer. Even with substantially more advanced technology, it would be almost impossible to examine every star in the Galaxy and obtain complete data for each one about its planets, their atmospheres, geological histories, and possible biospheres. For stars in other galaxies, such a task would be incomparably more demanding.

The scale of the problem can already be illustrated by a simple thought experiment. If we assume that the Milky Way contains at least 100 billion stars, then our successors, if they wished to physically examine every one of them within a thousand years, would have to investigate approximately 274,000 stars per day on average. Even the hypothetical possibility of travelling at or beyond the speed of light would solve only part of the problem. They would still need an enormous number of research devices, would have to coordinate their trajectories, provide energy, and devote enough time to observing each planetary system. A complete empirical survey of the Galaxy would therefore be extraordinarily difficult, and when it comes to the entire universe, such an undertaking is even harder to imagine.

Part of the answer to the apparent absence of other civilizations may already lie in these scales. If a planet with conditions suitable for the long-term preservation of life occurred around only one in several thousand stars, the number of interesting targets would quickly decrease. Astronomical habitability is, moreover, only the beginning of a long and poorly understood chain of events. We do not know the probability of the origin of life, the development of complex organisms, the emergence of intelligence, or the development of a technological civilization, nor do we know what proportion of such civilizations survive long enough to become interstellar. There could also be an additional civilizational selection filter, connected with the ability to limit violence, internal conflicts, and self-destructive potential, which increases together with technological power. Nor do we know for how long during its existence such a civilization remains detectable from afar.

Distances represent only one side of the problem. The other is time. Two civilizations may arise in the same Galaxy, yet be separated by a million years. One may disappear before the other even emerges. They never meet, and it is entirely possible that they never detect one another. The apparent silence of the universe therefore does not yet allow us to conclude that the universe is empty. The reason may lie in the rarity of life, immense distances, the short period during which technological civilizations are detectable, or simply in the limitations of our observational methods.

All of this raises an even deeper question. What would it actually mean for humanity to learn that millions of technological civilizations currently exist in the universe, perhaps including some that are more intelligent than we are? Such a discovery would undoubtedly represent one of the greatest epistemological, anthropological, and existential turning points in the history of humanity. It is difficult, however, to know what such knowledge would do to the human being. We would answer one of the oldest questions concerning our own existence and satisfy part of our desire for knowledge, while at the same time being forced to reconsider our conception of our own position in the universe. Humanity would become one among potentially millions of independent manifestations of consciousness and intelligence.

Human life itself would not thereby lose its value. What would change above all would be our conception of our own exceptional status. If we discovered that intelligence exists on millions of other worlds, we would be less rare as a phenomenon, but our lives would not thereby become any less real or important. The value of humanity is, after all, not determined solely by its rarity in the universe. It is also determined by our capacities for understanding, creating, preserving, and giving to others, as well as by the way in which we connect our knowledge with moral and ethical responsibility.

Perhaps, therefore, the most important result of such equations will not be a definitive answer to the question of how many hypothetical ExoEarth candidates exist, but rather a clearer recognition of the limits of our knowledge and of the responsibility we bear as inhabitants of what is currently the only planet with an active biosphere that we know with certainty exists. Behind life on Earth stand billions of years of successful evolutionary history, while its most recent result is a technological civilization whose own future, however, remains subject to very great uncertainty...